

COMP0078: Supervised Learning

Complete Problem Sheets with Solutions

(Sheets 1–7 + Supplementary Exam)

UCL Department of Computer Science

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1 Problem Sheet 1: Introduction to Supervised Learning

Solutions by Antonin Schrab & Lisa Tse

Problem 1. “For the K -nearest neighbour algorithm, larger K values will tend to lead to overfitting.” True or False? Explain your answer.

Solution. False. Larger k creates more regular, smoother decision boundaries, which could possibly lead to underfitting.

Problem 2. First, give Cover’s bound on the Bayes error of the 1-nearest neighbour algorithm as the number of examples goes to infinity. Second, give an example where the 1-nearest neighbour algorithm achieves the Bayes error as the number of examples goes to infinity, explain your reasoning.

Solution. Cover’s bound states that the 1-NN error L_{1-NN} satisfies

$$L^* \leq L_{1-NN} \leq 2L^*(1 - L^*)$$

where $L^* = 1 - \sum_c \max_c P(c|x)$ is the Bayes error (for the multiclass case with C classes).

Two possibilities where 1-NN achieves the Bayes error are when $p^* = 1/C$ (each class is equally likely) or $p^* = 1$ (deterministic case, when the Bayes error = 0). In the former case, $\sum_c P(c|x)(1 - P(c|x)) = 1 - \frac{1}{C}$ and $1 - p^* = 1 - \frac{1}{C}$. In the latter case $\sum_c P(c|x)(1 - P(c|x)) = 1 - p^* = 0$.

Problem 3. How can we choose k to ensure that the k -NN makes an unambiguous decision in the two-class case provided the test point is not equidistant to any pair of points in the training set? What values of k give an unambiguous decision under similar assumptions for the 3-class case?

Solution. 2-class case: k is odd (if $k = 2r$ for $r \in \mathbb{N} \setminus \{0\}$ is even then r points labelled from one class and the r other points labelled from the other class).

3-class case: $k = 1$ (if $k = 2r$ for $r \in \mathbb{N} \setminus \{0\}$ then the same reasoning as in the 2-class case applies; if $k = 2r + 1$ for $r \in \mathbb{N} \setminus \{0\}$ then we can have one point labelled from the third class and the $2r$ remaining points labelled as in the 2-class case).

Problem 4. One of the drawbacks of the nearest-neighbour algorithm is that we must retain all of the training data. Describe a situation where a training point can be removed without affecting the resulting 1-NN classification for any test point in the input space.

Solution. Start by constructing a Voronoi diagram, that is, for each point x we construct a region R_x such that every point $y \in R_x$ is closer to x than it is to all the other points.

Now, if a region R_x of a point x with label ℓ has boundaries only with regions of points with the same label ℓ , then we can remove the point x without affecting the resulting 1-NN classification for any test point in the input space. This is because, after having removed x , every point in the region R_x will be closest to one of the points of the regions sharing a boundary with R_x which has the same label.

Problem 5. In linear regression it is common to transform the training data

$$(x_1, y_1), \dots, (x_m, y_m)$$

to

$$((x_1, 1), y_1), \dots, ((x_m, 1), y_m),$$

i.e., we add an additional component to each input vector and set it to 1. What is the motivation for this procedure?

Solution. To incorporate a bias term.

Problem 6. Given we have a dataset $\{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$ where $x \in \mathbb{R}^n$ and $y \in \{1, 2, \dots, K\}$. Thus the inputs are real vectors and the labels denote one of K classes. How can linear regression (least squares) be used or adapted to tackle such a multi-class classification problem?

Solution. Using one-vs-one (OvO) or one-vs-rest (OvR) strategies. For OvR, there are K classifiers and each binary classifier is fed $+1$ for a class $k \in [K]$, and -1 for all the other classes. It combines the predictions by predicting the class with the highest score $\langle w, x_i \rangle - \theta$. For OvO, each classifier is fed one of the possible combinations of two of the K classes, and predicts according to a majority vote.

Problem 7. Given $\mathcal{Y} \subset \mathbb{R}$ and $|\mathcal{Y}| < \infty$ and $|\mathcal{X}| < \infty$ and a probability mass function $p : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, find

$$f^* := \operatorname{argmin}_f \mathbb{E}[|y - f(x)|]$$

(Bayes estimator with L_1 loss).

Solution. Let $\mathcal{Y} \subseteq \mathbb{R}$ and $L(y, \hat{y}) := |y - \hat{y}|$. The Bayes estimator is

$$f^* := \operatorname{argmin}_f \mathbb{E}[L(y, f(x))] = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} |y - f(x)| p(x, y) = \sum_{x \in \mathcal{X}} \left(\sum_{y \in \mathcal{Y}} |y - f(x)| p(y|x) \right) p(x).$$

As a function of x , we let $\hat{y} := f^*(x)$. Now, set $x = x_0$. We then have

$$\sum_{y \in \mathcal{Y}} |y - f(x_0)| p(y|x_0) p(x_0) \propto \sum_{y \in \mathcal{Y}} |y - f(x_0)| p(y|x_0)$$

as $p(x_0)$ does not depend on y . Since this is convex, to find the minimum we take the derivative with respect to $\hat{y}$ and set it equal to 0:

$$\begin{aligned} 0 &= \frac{\partial}{\partial \hat{y}} \sum_{y \in \mathcal{Y}} |y - \hat{y}| p(y|x_0) \\ 0 &= \frac{\partial}{\partial \hat{y}} \sum_{\substack{y \in \mathcal{Y} \\ y \geq \hat{y}}} (y - \hat{y}) p(y|x_0) + \frac{\partial}{\partial \hat{y}} \sum_{\substack{y \in \mathcal{Y} \\ y < \hat{y}}} (\hat{y} - y) p(y|x_0) \\ 0 &= \sum_{\substack{y \in \mathcal{Y} \\ y \geq \hat{y}}} (-1) \cdot p(y|x_0) + \sum_{\substack{y \in \mathcal{Y} \\ y < \hat{y}}} 1 \cdot p(y|x_0) \\ 0 &= -P(Y \geq \hat{y}|X = x_0) + P(Y < \hat{y}|X = x_0). \end{aligned}$$

Hence

$$P(Y \geq \hat{y}|X = x_0) = P(Y < \hat{y}|X = x_0) = 1 - P(Y \geq \hat{y}|X = x_0) \implies P(Y \geq \hat{y}|X = x_0) = \frac{1}{2}.$$

By definition, a median of a real random variable Z is a value $m \in \mathbb{R}$ satisfying $P(Z \leq m) \geq \frac{1}{2}$ and $P(Z \geq m) \geq \frac{1}{2}$.

We have $P(Y \geq \hat{y}|X = x_0) = \frac{1}{2}$ and $P(Y \leq \hat{y}|X = x_0) \geq P(Y < \hat{y}|X = x_0) = \frac{1}{2}$, so we conclude that the Bayes estimator $\hat{y}(x)$ is the **median** of $Y|X = x$.

2 Problem Sheet 2: Features, Kernels and Regularisation

Solutions by Antonin Schrab

2.1 Convexity

Problem 8 (1.1.1). *If f and g are convex then $f + g$ is convex.*

Solution. For $p, q \in \mathcal{X}$ and $\alpha \in (0, 1)$, we have

$$\begin{aligned}(f + g)(\alpha p + (1 - \alpha)q) &= f(\alpha p + (1 - \alpha)q) + g(\alpha p + (1 - \alpha)q) \\ &\leq \alpha f(p) + (1 - \alpha)f(q) + \alpha g(p) + (1 - \alpha)g(q) \\ &= \alpha(f(p) + g(p)) + (1 - \alpha)(f(q) + g(q)).\end{aligned}$$

This shows that $f + g$ is convex.

Problem 9 (1.1.2). *If f is convex and g is affine (linear + a constant) then $f(g(\cdot))$ is convex.*

Solution. Since g is affine it can be written as $g(x) = a(x + b) = ax + ab$ for some $a, b \in \mathbb{R}$ and for all $x \in \mathbb{R}$. We then have

$$\begin{aligned}(f \circ g)(\alpha p + (1 - \alpha)q) &= f(g(\alpha p + (1 - \alpha)q)) \\ &= f(\alpha ap + (1 - \alpha)aq + ab) \\ &= f(\alpha a(p + b) + (1 - \alpha)a(q + b)) \\ &\leq \alpha f(a(p + b)) + (1 - \alpha)f(a(q + b)) \\ &= \alpha f(g(p)) + (1 - \alpha)f(g(q)).\end{aligned}$$

This shows that $f \circ g$ is convex.

Problem 10 (1.1.3). *Suppose M is a symmetric matrix then M is a PSD matrix if and only if $f(x) = x^\top Mx$ is convex.*

Solution. Suppose M is a symmetric PSD matrix. Note that the Hessian of f is

$$\frac{\partial^2}{\partial x^2} f(x) = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} x^\top Mx \right) = \frac{\partial}{\partial x} (M + M^\top)x = \frac{\partial}{\partial x} 2Mx = 2M,$$

which is PSD since M is PSD. It follows from Problem 1.1.5 (proved below) that f is convex.

Now, suppose f is convex and so it attains its minimum at

$$0 = \frac{\partial}{\partial x} f(x) = \frac{\partial}{\partial x} x^\top Mx = (M + M^\top)x = 2Mx.$$

We deduce that f achieves its minimum at 0, so

$$x^\top Mx = f(x) \geq f(0) = 0^\top M0 = 0.$$

This shows that M is positive semidefinite.

Problem 11 (1.1.4). *A level set of a convex function is convex.*

Solution. Let f be a convex function. The level set of f at c is $L := \{x : f(x) \leq c\}$. Suppose $p, q \in L$, that is, $f(p) \leq c$ and $f(q) \leq c$. We then have

$$f(\alpha p + (1 - \alpha)q) \leq \alpha f(p) + (1 - \alpha)f(q) \leq \alpha c + (1 - \alpha)c = c,$$

so $\alpha p + (1 - \alpha)q \in L$. This proves that the level set L of f is a convex set.

Problem 12 (1.1.5). For $f : \mathcal{X} \rightarrow \mathbb{R}$, if $\mathcal{X} \subseteq \mathbb{R}^n$ is a convex set and if the Hessian of f evaluated at x denoted $H(x)$ is positive semidefinite for all $x \in \mathcal{X}$, then f is convex.

Solution. Let $x, a \in \mathcal{X}$. By Taylor's Theorem, there exists $t \in (0, 1)$ such that

$$f(x) = f(a) + (x - a)^\top \nabla f(a) + \frac{1}{2}(x - a)^\top H(a + t(x - a))(x - a).$$

Since $H(a + t(x - a))$ is PSD by assumption, for $x - a \in \mathcal{X}$, we have

$$(x - a)^\top H(a + t(x - a))(x - a) \geq 0,$$

from which we deduce that $f(x) \geq f(a) + (x - a)^\top \nabla f(a)$.

Now, consider $p, q \in \mathcal{X}$, $\alpha \in [0, 1]$, and let $a := \alpha p + (1 - \alpha)q$ which belongs to $\mathcal{X}$ as $\mathcal{X}$ is a convex set. Note that $p - a = (1 - \alpha)(p - q)$ and $q - a = \alpha(q - p)$. We have

$$\begin{aligned} \alpha f(p) + (1 - \alpha)f(q) &\geq \alpha \left(f(a) + (p - a)^\top \nabla f(a) \right) + (1 - \alpha) \left(f(a) + (q - a)^\top \nabla f(a) \right) \\ &= f(a) + \nabla f(a)^\top (\alpha(p - a) + (1 - \alpha)(q - a)) \\ &= f(a) + \nabla f(a)^\top (\alpha(1 - \alpha)(p - q) + (1 - \alpha)\alpha(q - p)) \\ &= f(a) = f(\alpha p + (1 - \alpha)q), \end{aligned}$$

which proves that f is convex.

2.2 Primal vs Dual

Problem 13 (1.2). Consider the solution to the linear regression optimisation problem. When is it advantageous to compute it via the primal solution? When is it advantageous to compute it via the dual solution? Explain why.

Solution. Recall that the matrix X is of size $m \times n$ where we have m samples in $\mathbb{R}^n$. The primal solution is

$$w = (X^\top X + \lambda I_n)^{-1} X^\top y$$

and the dual solution is

$$w = \sum_{i=1}^m \alpha_i x_i \quad \text{where} \quad \alpha = (XX^\top + \lambda I_m)^{-1} y.$$

The operation with the highest time complexity in those formulas are the two matrix inversions. If we have more dimensions n than samples m , it is therefore advantageous to compute via the dual solution. For the case where we have more samples m than dimensions n , it is advantageous to use the primal solution.

2.3 Kernel Derivations

Problem 14 (1.3). Given a kernel $K : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ where $K(x, t) := (1 + \langle x, t \rangle)^2$. Find a feature map $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^6$ which corresponds to the kernel.

Solution. We have

$$\begin{aligned} K(x, t) &= (1 + \langle x, t \rangle)^2 = (1 + x_1 t_1 + x_2 t_2)^2 \\ &= 1 + x_1^2 t_1^2 + x_2^2 t_2^2 + 2x_1 t_1 + 2x_2 t_2 + 2x_1 x_2 t_1 t_2 \\ &= \begin{pmatrix} 1 \\ x_1^2 \\ x_2^2 \\ \sqrt{2} x_1 \\ \sqrt{2} x_2 \\ \sqrt{2} x_1 x_2 \end{pmatrix}^\top \begin{pmatrix} 1 \\ t_1^2 \\ t_2^2 \\ \sqrt{2} t_1 \\ \sqrt{2} t_2 \\ \sqrt{2} t_1 t_2 \end{pmatrix} =: \varphi(x)^\top \varphi(t). \end{aligned}$$

Problem 15 (1.4). For each of the following functions $K : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ argue whether they are a valid kernel (i.e. the kernel can be written as an inner product in some feature space) and when the answer is positive derive an associated feature map representation.

1. $K(x, t) = x^\top D t$ where $D := \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$

2. $K(x, t) = x^\top D t$ where $D := \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix}$

3. $K(x, t) = \exp(x_1 t_1)$ where x_1 is the first component of x and t_1 is the first component of t .

4. $K(x, t) = x^\top t - (x^\top t)^2$

5. Now prove that if $K : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is a given valid kernel then the following transformed kernels are also valid:

(a) $K(Ax, At)$ where A is a given 2×2 matrix.

(b) $f(x)K(x, t)f(t)$ where f is a given real-valued function.

Solution. 1.4.1: This is a kernel and the feature map can be obtained as

$$K(x, t) = x^\top D t = x_1 t_1 + 5x_2 t_2 = \begin{pmatrix} x_1 \\ \sqrt{5}x_2 \end{pmatrix}^\top \begin{pmatrix} t_1 \\ \sqrt{5}t_2 \end{pmatrix} =: \varphi(x)^\top \varphi(t).$$

1.4.2: Note that, for $x = (1, 0)^\top \in \mathbb{R}^2$ and $c = 1 \in \mathbb{R}$, we have

$$c^2 K(x, x) = (1 \ 0) \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = -1 < 0.$$

So the function K is not positive semidefinite, hence, it does not define a valid kernel.

1.4.3: Using the Taylor series of the exponential function $\exp(u) = \sum_{n=0}^{\infty} \frac{u^n}{n!}$, we obtain

$$K(x, t) = \exp(x_1 t_1) = \sum_{n=0}^{\infty} \frac{x_1^n t_1^n}{n!} = \begin{pmatrix} 1 \\ x_1 \\ x_1^2/\sqrt{2!} \\ x_1^3/\sqrt{3!} \\ \vdots \end{pmatrix}^\top \begin{pmatrix} 1 \\ t_1 \\ t_1^2/\sqrt{2!} \\ t_1^3/\sqrt{3!} \\ \vdots \end{pmatrix} =: \varphi(x)^\top \varphi(t).$$

We deduce that K is a valid kernel.

1.4.4: Note that, for $x = (\sqrt{2}, 0)^\top \in \mathbb{R}^2$ and $c = 1 \in \mathbb{R}$, we have

$$c^2 K(x, x) = 2 - 4 = -2 < 0.$$

So the function K is not positive semidefinite, hence, it does not define a valid kernel.

1.4.5(a): Consider $c_1, \dots, c_m \in \mathbb{R}$ and $x_1, \dots, x_m \in \mathbb{R}^2$. Letting $\tilde{x}_i := Ax_i \in \mathbb{R}^2$ for $i = 1, \dots, m$. Since K is a kernel, we have

$$\sum_{i=1}^m \sum_{j=1}^m c_i c_j K(\tilde{x}_i, \tilde{x}_j) \geq 0,$$

or equivalently $\sum_{i=1}^m \sum_{j=1}^m c_i c_j K(Ax_i, Ax_j) \geq 0$. This proves that $K(Ax, At)$ is positive semidefinite and hence it is a valid kernel.

1.4.5(b): Consider $c_1, \dots, c_m \in \mathbb{R}$ and $x_1, \dots, x_m \in \mathbb{R}^2$. Let $\tilde{c}_i := f(x_i)c_i \in \mathbb{R}$ for $i = 1, \dots, m$. Since K is a kernel, we have

$$\sum_{i=1}^m \sum_{j=1}^m \tilde{c}_i \tilde{c}_j K(x_i, x_j) \geq 0,$$

or equivalently $\sum_{i=1}^m \sum_{j=1}^m c_i c_j f(x_i) K(x_i, x_j) f(x_j) \geq 0$. This proves that $f(x)K(x, t)f(t)$ is positive semidefinite and hence it is a valid kernel.

Problem 16 (1.5). Consider a Gaussian kernel function $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $K(x, z) := e^{-\|x-z\|^2}$. Does there exist a finite-dimensional feature map representation? I.e., does there exist a $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^d$ such that $K(x, z) = \langle \varphi(x), \varphi(z) \rangle$? Indicate an answer “yes” or “no” and provide an argument supporting your answer. [hard]

Solution. No.

Suppose for contradiction that there exists a finite-dimensional feature map representation, that is there exists $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^d$ for some finite $d \in \mathbb{N}$ such that

$$e^{-\|x-z\|^2} = K(x, z) = \sum_{i=1}^d \varphi_i(x) \varphi_i(z).$$

Consider the points $x_\ell := (\ell, 0, \dots, 0) \in \mathbb{R}^n$ for $\ell \in \mathbb{N} \setminus \{0\}$. For $\ell \neq h$, the angle θ between $\varphi(x_\ell)$ and $\varphi(x_h)$ in $\mathbb{R}^d$ satisfies

$$\cos \theta = \frac{\varphi(x_\ell)^\top \varphi(x_h)}{(\varphi(x_\ell)^\top \varphi(x_\ell))(\varphi(x_h)^\top \varphi(x_h))} = \frac{K(x_\ell, x_h)}{K(x_\ell, x_\ell)K(x_h, x_h)} = e^{-|\ell-h|} \leq e^{-1}.$$

Since arccos is a decreasing function on $[-1, 1]$, we obtain that $\theta \geq \arccos(e^{-1}) > 0$.

This means that we have infinitely many vectors $\{\varphi(x_\ell) : \ell \in \mathbb{N} \setminus \{0\}\}$ in $\mathbb{R}^d$ with the angle between any pair of these being greater than $e^{-1} > 0$. This is a contradiction as there exist only finitely many such vectors in $\mathbb{R}^d$. Having reached a contradiction, we deduce that there does not exist such a finite-dimensional feature map.

2.4 Vector Spaces and Kernels

Problem 17 (2.1). Argue that the vector space definition implies that every element $x \in \mathcal{X}$ has an additive inverse $-x \in \mathcal{X}$ such that $x + (-x) = 0$.

Solution. We have

$$0 = 0x = (1 - 1)x = 1x + (-1)x = x + (-1)x.$$

We deduce that the additive inverse of $x \in \mathcal{X}$ is $(-1)x \in \mathcal{X}$ which we simply write as $-x \in \mathcal{X}$.

Problem 18 (2.2). Kernels between sets. Let $\mathcal{X}$ be a finite set. Define $K : 2^{\mathcal{X}} \times 2^{\mathcal{X}} \rightarrow \mathbb{R}$ as

$$K(A, B) := 2^{|A \cap B|}$$

where $A, B \subseteq \mathcal{X}$. Prove that K is a kernel.

Solution. Recall that $2^{\mathcal{X}}$ is the power set of $\mathcal{X}$ (i.e. the set of all subsets of $\mathcal{X}$, including the empty set), and that $|2^{\mathcal{X}}| = 2^{|\mathcal{X}|}$.

To prove that K is a kernel, we construct a feature map. For a statement S which is either true or false, we use the notation $[S] = 1$ if S is true and $[S] = 0$ if S is false.

Now, define the features $\varphi_C(A) = [C \subseteq A]$ indexed by subsets $C \in 2^{\mathcal{X}}$. For $A, B \in 2^{\mathcal{X}}$, we then have

$$\sum_{C \in 2^{\mathcal{X}}} \varphi_C(A) \varphi_C(B) = \sum_{C \in 2^{\mathcal{X}}} [C \subseteq A][C \subseteq B] = \sum_{C \in 2^{\mathcal{X}}} [C \subseteq A \cap B] = \sum_{C \in 2^{A \cap B}} 1 = |2^{A \cap B}| = 2^{|A \cap B|} = K(A, B).$$

This proves that K is indeed a kernel.

Problem 19 (2.3). *Min kernel.*

1. Argue that $\min(x, t)$ (where $x, t \in [0, \infty)$) is a kernel. **[technical]**
2. Let $M > 0$. Find a set of basis functions $\varphi_\ell : [0, M] \rightarrow \mathbb{R}$ for $\ell \in \mathbb{N}$ such that $\min(x, t) = \sum_{\ell=0}^{\infty} \varphi_\ell(x) \varphi_\ell(t)$. **[technical, very difficult]**

Solution. 2.3.1: Consider the inner product space $\mathcal{F}$ which consists of functions from $[0, \infty)$ to $\mathbb{R}$ satisfying: (1) $f(0) = 0$, (2) f is absolutely continuous (hence $f(b) - f(a) = \int_a^b f'(u) du$), (3) $\int_0^\infty (f'(u))^2 du < \infty$, with the inner product

$$\langle f, g \rangle := \int_0^\infty f'(u) g'(u) du$$

for $f, g \in \mathcal{F}$.

We consider the feature map $\varphi : [0, \infty) \rightarrow \mathcal{F}$ defined as $\varphi(x) := \Phi_x \in \mathcal{F}$ for all $x \geq 0$, where

$$\Phi_x(u) := \begin{cases} u & \text{if } u \leq x, \\ x & \text{if } u > x, \end{cases}$$

for all $u \geq 0$. When we differentiate Φ_x we obtain the step-function

$$\Phi'_x(u) = \begin{cases} 1 & \text{if } u \leq x, \\ 0 & \text{if } u > x. \end{cases}$$

Note that, for $x, t \in \mathbb{R}$, we have

$$\Phi'_x(u) \Phi'_t(u) = \begin{cases} 1 & \text{if } u \leq \min(x, t), \\ 0 & \text{if } u > \min(x, t). \end{cases}$$

Finally, for $x, t \in \mathbb{R}$, we have

$$\langle \varphi(x), \varphi(t) \rangle = \langle \Phi_x, \Phi_t \rangle = \int_0^\infty \Phi'_x(u) \Phi'_t(u) du = \int_0^{\min(x, t)} 1 du + \int_{\min(x, t)}^\infty 0 du = \min(x, t).$$

This proves that $\min(x, t)$ is a kernel.

2.3.2: We work with the kernel $k(x, t) := \min(x, t)$ for $x, t \in [0, M]$. Consider the integral operator $T_k : L^2[0, M] \rightarrow L^2[0, M]$ which for $f \in L^2[0, M]$, is defined as $(T_k f)(x) := \int_0^M k(x, u) f(u) du$. Mercer's Theorem states that

$$k(x, t) = \sum_{\ell=0}^{\infty} \lambda_\ell e_\ell(x) e_\ell(t)$$

where $\{e_\ell\}_{\ell=0}^\infty$ is an orthonormal basis for $L^2[0, M]$ consisting of eigenfunctions of T_k with associated eigenvalues $\{\lambda_\ell\}_{\ell=0}^\infty$.

An eigenpair (λ, e) satisfies for all $x \in [0, M]$:

$$\lambda e(x) = \int_0^x t e(t) dt + \int_x^M x e(t) dt.$$

Setting $x = 0$ gives $\lambda e(0) = 0$, so $e(0) = 0$. Differentiating using Leibniz rule gives $\lambda e'(x) = \int_x^M e(t) dt$. Setting $x = M$ gives $\lambda e'(M) = 0$, so $e'(M) = 0$. Differentiating again gives $\lambda e''(x) = -e(x)$, which is a second-order ODE with solution

$$e(x) = c_1 \cos\left(\frac{x}{\sqrt{\lambda}}\right) + c_2 \sin\left(\frac{x}{\sqrt{\lambda}}\right).$$

From $c_1 = e(0) = 0$ and $0 = e'(M) = \frac{c_2}{\sqrt{\lambda}} \cos\left(\frac{M}{\sqrt{\lambda}}\right)$, we get

$$\frac{M}{\sqrt{\lambda_\ell}} = \left(\ell + \frac{1}{2}\right) \pi \iff \lambda_\ell = \frac{M^2}{\left(\ell + \frac{1}{2}\right)^2 \pi^2}, \quad \ell \in \mathbb{N}.$$

Normalizing $\|e_\ell\|_2^2 = 1$ gives $c_2 = \sqrt{2/M}$. Thus the feature map is

$$\varphi_\ell(x) := \sqrt{\lambda_\ell} e_\ell(x) = \frac{\sqrt{2} M}{\left(\ell + \frac{1}{2}\right) \pi} \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right).$$

3 Problem Sheet 3: Support Vector Machines

Solutions by Stephen Pasteris & Antonin Schrab

Problem 20 (1.1). Describe the criterion used by hard margin Support Vector Machines to choose a separating hyperplane for a linearly separable dataset, illustrating with a diagram indicating the margin and the support vectors.

Solution. The criterion is to choose the separating hyperplane (i.e. an affine plane which has all positive examples on one side and all negative examples on the other) that maximises the distance of the nearest example to it.

Problem 21 (1.2). For a hard-margin SVM. If we remove one of:

1. the examples which is a support vector from the training set,
2. the examples which is not a support vector from the training set,

and retrain without that example. How does the maximum margin change?

Solution. If we remove one example which is a support vector (1.2.1), then either the maximum margin remains the same or it increases; the margin never decreases.

If we remove examples which are not support vectors (1.2.2), then the optimal hyperplane with its maximum margin remains the same.

Problem 22 (1.3). Consider the optimisation given by

$$\min_{w, \xi} \|w\| + C \sum_{i=1}^m \xi_i \quad \text{subject to } y_i \langle w, x_i \rangle \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \dots, m.$$

Which part of the optimisation corresponds to the regulariser? What is the loss function incorporated into the optimisation?

Solution. We can rewrite the optimisation problem as

$$\min_w \sum_{i=1}^m \max(0, 1 - y_i \langle w, x_i \rangle) + C^{-1} \|w\|,$$

that is

$$\min_w \sum_{i=1}^m V_{\text{hinge}}(y_i, \langle w, x_i \rangle) + C^{-1} \|w\|.$$

So, the loss function incorporated into the optimisation is the **hinge loss**, the term $\|w\|$ corresponds to the **regulariser** with regularisation coefficient C^{-1} .

Problem 23 (2.1). Assume that the set $S = \{(x_i, y_i)\}_{i=1}^m \subset \mathbb{R}^2 \times \{-1, 1\}$ of binary examples is strictly linearly separable by a line going through the origin, that is, there exists a vector $w \in \mathbb{R}^2$ such that the linear function $f(x) = w^\top x$, $x \in \mathbb{R}^2$ has the property that $y_i f(x_i) > 0$ for every $i = 1, \dots, m$. Consider the optimisation problem (linearly separable SVM):

$$\mathbf{P1:} \quad \min_w \frac{1}{2} w^\top w \quad \text{subject to } y_i w^\top x_i \geq 1, \quad i = 1, \dots, m.$$

Argue that the above problem has a unique solution. Describe the geometric meaning of this solution.

Solution. Solution exists. Since the data is strictly linearly separable by a line going through the origin (by assumption), there exists some $\bar{w} \in \mathbb{R}^2$ satisfying $y_i \bar{w}^\top x_i \geq 1$ for $i = 1, \dots, m$. Hence, the optimisation problem reduces to optimization over a compact set, and a solution exists.

Solution is unique. The function $\frac{1}{2}w^\top w$ is strictly convex since $\frac{\partial^2}{\partial w^2}(\frac{1}{2}w^\top w) = I > 0$. The constraint set is a convex (non-empty) compact set (intersection of half-spaces with a ball). Our problem is then to minimise a strictly convex function over a non-empty convex compact set, which always has a unique solution.

Geometric meaning. The geometric meaning of the solution is that it is the separating hyperplane, going through the origin, of which the distance of the nearest example to it is maximised.

Problem 24 (2.2). Show that the vector w solving problem P1 has the form $w = \sum_{i=1}^m c_i y_i x_i$ where $c_1, \dots, c_m$ are some nonnegative coefficients. [HINT: use the method of Lagrange multipliers]

Solution. Using the method of Lagrange multipliers, the optimisation problem P1 is equivalent to

$$P1^* : \max_{c \geq 0} \min_w L(w, c)$$

where the Lagrangian is

$$L(w, c) = \frac{1}{2}w^\top w - \sum_{i=1}^m c_i (y_i w^\top x_i - 1).$$

We first solve $\min_w L(w, c)$ by solving

$$0 = \nabla_w L = w - \sum_{i=1}^m c_i y_i x_i,$$

which gives us $w = \sum_{i=1}^m c_i y_i x_i$.

Problem 25 (2.3). Show that the coefficients $c_1, \dots, c_m$ in the above formula solve the optimization problem

$$P2: \max_{c \geq 0} -\frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m c_i c_j y_i y_j x_i^\top x_j + \sum_{i=1}^m c_i.$$

Finally, if $(c_1, \dots, c_m)$ is the solution to this problem and w is the solution to problem P1, argue that $w^\top w = \sum_{i=1}^m c_i$.

Solution. From Solution 2.2, for $w = \sum_{i=1}^m c_i y_i x_i$, we have

$$\begin{aligned} L(w, c) &= \frac{1}{2}w^\top w - \sum_{i=1}^m c_i y_i w^\top x_i + \sum_{i=1}^m c_i \\ &= \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m c_i c_j y_i y_j x_i^\top x_j - \sum_{i=1}^m \sum_{j=1}^m c_i c_j y_i y_j x_i^\top x_j + \sum_{i=1}^m c_i \\ &= -\frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m c_i c_j y_i y_j x_i^\top x_j + \sum_{i=1}^m c_i. \end{aligned}$$

This gives the dual problem P2. To show $w^\top w = \sum_{i=1}^m c_i$: by stationarity, for i with $c_i > 0$, we have $\sum_{j=1}^m c_j y_i y_j x_i^\top x_j = 1$. Hence,

$$w^\top w = \sum_{i=1}^m \sum_{j=1}^m c_i c_j y_i y_j x_i^\top x_j = \sum_{i: c_i > 0} c_i \cdot 1 = \sum_{i=1}^m c_i.$$

4 Problem Sheet 4: Tree-based Learning and Ensemble Methods

Solutions by Shota Saito & Antonin Schrab

Problem 26 (1.1). Given the dataset $\{((1,1),9),((1,2),-4),((1,3),2),((2,2),4),((2,3),2)\}$, find the regression tree corresponding to the greedy recursive partitioning procedure with respect to square error.

Solution. Each element of the dataset is of the form (x,y) where the point $x = (x_1,x_2)$ has label y .

First partition A. The possible partitions are: $A_1: x_2 \leq 1.5$; $A_2: x_2 \leq 2.5$; $A_3: x_1 \leq 1.5$. Computing the square errors:

- A_1 : $R_1 = \{x : x_2 \leq 1.5\}$ with average $c_1 = 9$, $R_2 = \{x : x_2 > 1.5\}$ with average $c_2 = 1$. Error = $0 + 25 + 1 + 9 + 1 = 36$.
- A_2 : Error = 86.
- A_3 : Error ≈ 86.7 .

We pick A_1 with the smallest error.

Second partition B (within $\{x : x_2 > 1.5\}$): $B_1: x_2 \leq 2.5$ (error 32); $B_2: x_1 \leq 1.5$ (error 20). We pick B_2 .

Third and fourth partitions are each uniquely determined by the remaining two-point regions. The final tree is:

$x_2 \leq 1.5? \rightarrow$ Yes: predict 9. No: $x_1 \leq 1.5? \rightarrow$ Yes: $x_2 \leq 2.5? \rightarrow$ Yes: -4, No: 2. No: $x_2 \leq 2.5? \rightarrow$ Yes: 4, No: 2.

Problem 27 (1.2). Explain how a random forest is constructed. What is the motivation for the tree construction method?

Solution. The random forest method is designed to encourage each decision tree component to be uncorrelated ("wisdom of crowds"):

1. Draw M samples from the training data (sample with replacement).
2. Consider a subset of the features ($n_{\text{tree}} < n$ where n is the total number of features — typically $n_{\text{tree}} = \sqrt{n}$ or $\log(n)$).
3. Build a decision tree.
4. Repeat 1–3 N times to generate N trees.
5. For regression: predict the mean of the outputs. For classification: predict the majority vote.

Both bootstrap sampling and random feature selection contribute to decorrelating each tree. Decision trees are typically high variance, so are ideal for bagging (which reduces the variance).

Problem 28 (1.3). Both bagging and boosting produce predictions by creating an ensemble of functions. Explain how these ensembles differ.

Solution. **Bagging** considers each sample to be of equal weight but each learner only sees a subset of the data. The bias of each learner is the same as the ensemble. Our motivation is variance reduction.

Boosting places more weight on samples that previous models got wrong. Each learner sees all of the data but with variable sample weights. Learners are added in a non-i.i.d. way to remove bias.

Problem 29 (2.1). *The Adaboost initialisation and update.*

1. Explain how Adaboost chooses its initial weighting $D_0(i)$ of the examples $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$.
2. Explain how it updates the distribution D_t after choosing a weak learner h_t with coefficient α_t at stage t including the normalising constant Z_t .
3. Give an intuitive justification for this update.

Solution. 2.1.1: It chooses an equal weighting for all examples, i.e. $D_0(i) = 1/m$ for all i .

2.1.2: The update is given by

$$D_{t+1}(i) = \frac{D_t(i)e^{-\alpha_t y_i h_t(x_i)}}{Z_t}$$

where $\alpha_t := \frac{1}{2} \log\left(\frac{1-\epsilon_t}{\epsilon_t}\right)$, $\epsilon_t := \sum_{i=1}^m D_t(i) \mathbf{1}[h_t(x_i) \neq y_i]$ and $Z_t := \sum_{i=1}^m D_t(i) e^{-\alpha_t y_i h_t(x_i)}$.

2.1.3: If we have a correctly classified example, $y_i = h_t(x_i)$, so $y_i h_t(x_i) = 1$ and then $D_{t+1}(i) \propto D_t(i) e^{-\alpha_t}$. Since $\alpha_t \geq 0$, the weight for that example is reduced. If we have a wrongly classified example $y_i \neq h_t(x_i)$ so that $y_i h_t(x_i) = -1$, then $D_{t+1}(i) \propto D_t(i) e^{\alpha_t}$, i.e., the weight for that example is increased.

Problem 30 (2.2). *Explain how Adaboost uses the distribution D_t to choose a weak learner from a set $\mathcal{H}$ at stage t .*

Solution. Any hypothesis that satisfies $\epsilon_t := \sum_{i=1}^m D_t(i) \mathbf{1}[h_t(x_i) \neq y_i] < \frac{1}{2}$ will work. However, the optimal weak learner is the hypothesis that minimises ϵ_t .

Problem 31 (2.3). *Using the results of problem 2.1.2, obtain a bound for*

$$\frac{1}{m} \sum_{i=1}^m \mathbf{1}[\text{sign}(f_T(x_i)) \neq y_i] \leq \frac{1}{m} \sum_{i=1}^m \exp(-y_i f_T(x_i)),$$

in terms of the normalisation constants $Z_1, \dots, Z_T$ where $f_T(x) = \sum_{t=1}^T \alpha_t h_t(x)$.

Solution. First, we prove the inequality by showing $\mathbf{1}[\text{sign}(f_T(x_i)) \neq y_i] \leq \exp(-y_i f_T(x_i))$ for $i = 1, \dots, m$. If $\text{sign}(f_T(x_i)) = y_i$, then $\mathbf{1}[\cdot] = 0 \leq \exp(-y_i f_T(x_i))$. If $\text{sign}(f_T(x_i)) \neq y_i$, then $y_i f_T(x_i) \leq 0$ and so $\exp(-y_i f_T(x_i)) \geq 1 = \mathbf{1}[\cdot]$.

Now, from the update rule:

$$D_{T+1}(i) = \frac{1}{m} \cdot \frac{\prod_{t=1}^T e^{-\alpha_t y_i h_t(x_i)}}{\prod_{t=1}^T Z_t}.$$

We can thus write

$$\frac{1}{m} \sum_{i=1}^m \exp(-y_i f_T(x_i)) = \frac{1}{m} \sum_{i=1}^m \prod_{t=1}^T \exp(-y_i \alpha_t h_t(x_i)) = \sum_{i=1}^m D_{T+1}(i) \prod_{t=1}^T Z_t = \prod_{t=1}^T Z_t.$$

Problem 32 (3). *A dataset $(x_1, y_1), \dots, (x_m, y_m)$ is generic iff $x_i = x_j \implies y_i = y_j$. A set of functions $\mathcal{H}$ is called weakly-universal if for every generic dataset and weighting of the data $\{D_i\}_{i=1}^m$ such that $\sum_{i=1}^m D_i = 1$, there exists a function $h \in \mathcal{H}$ with weighted error $\sum_{i=1}^m D_i [h(x_i) \neq y_i] < 0.5$.*

1. A positive decision stump $h_{i,z} : \mathbb{R}^n \rightarrow \{-1, 1\}$ is defined as $h_{i,z}(x) := 1$ if $x_i \leq z$, -1 if $x_i > z$. The set of all decision stumps is $\mathcal{H}_{ds} := \{s \cdot h_{i,z} : s \in \{-1, 1\}, i \in \{1, \dots, n\}, z \in \mathbb{R}\}$. Is $\mathcal{H}_{ds}$ weakly-universal?

2. Let $\mathcal{S}$ denote the set of all strings of non-zero finite length over $\{a, b\}$. A positive substring-match function $g_z : \mathcal{S} \rightarrow \{-1, 1\}$ is defined as $g_z(x) := 1$ if $x \sqsubseteq z$, -1 if $x \not\sqsubseteq z$. The set of all substring-match functions is $\mathcal{H}_{SS} := \{s \cdot g_z : s \in \{-1, 1\}, z \in \mathcal{S}\}$. Is $\mathcal{H}_{SS}$ weakly-universal?

Solution. 3.1: $\mathcal{H}_{ds}^1$ is weakly-universal for $n = 1$, but $\mathcal{H}_{ds}$ is not weakly-universal for $n > 1$.

For $n = 1$: WLOG assume $x_1 < x_2 < \dots < x_m$. Take $z = x_m$ so h_{1,x_m} predicts 1 for all points. If the weighted error is < 0.5 , done. If > 0.5 , use $-h_{1,x_m}$. If exactly 0.5, take $z = \frac{x_{m-1} + x_m}{2}$ and check: depending on y_m , either $h_{1,z}$ or $-h_{1,z}$ has weighted error < 0.5 (using $D_m > 0$).

For $n > 1$: Counter-example with the XOR dataset: $x_{0,0} = (0, 0, \dots)$ with $y = -1$, $x_{0,1} = (0, 1, 0, \dots)$ with $y = 1$, $x_{1,0} = (1, 0, 0, \dots)$ with $y = 1$, $x_{1,1} = (1, 1, 0, \dots)$ with $y = -1$, with uniform weights $D_{i,j} = 0.25$. Every decision stump in $\mathcal{H}_{ds}$ classifies exactly 2 points correctly and 2 wrongly, giving weighted error exactly 0.5.

3.2: $\mathcal{H}_{SS}$ is weakly-universal. The proof follows the same structure as for $n = 1$ decision stumps: take $z = x_1 \dots x_m$ (concatenation of all strings), so g_z predicts 1 for all. If error = 0.5, remove the longest string x_k from the concatenation to construct z' such that $x_i \sqsubseteq z'$ for $i \neq k$ but $x_k \not\sqsubseteq z'$, and the error shifts by $\pm D_k$ away from 0.5.

5 Problem Sheet 5: Online Learning I

Solutions by Lisa Tse and Ang Ming Liang

Problem 33 (1). Suppose $\mathcal{X} = \{\text{True}, \text{False}\}^n$. Give a polynomial time algorithm A with a mistake bound $M(A) \leq O(n^2)$ for any example sequence which is consistent with a k -literal conjunction.

Solution. We take the perceptron algorithm. We have the mistake bound $M \leq R^2 \|u\|^2$ where $R^2 = \max_t \|x_t\|^2$, and this holds for any u for which $\langle u, x_t \rangle y_t \geq 1$. Using de Morgan's Law to represent the conjunction as a disjunction, define

$$u_i = \begin{cases} -2 & \text{if } i \in S \\ 1 & \text{if } i = n+1 \\ 0 & \text{otherwise} \end{cases}$$

and the feature map $\varphi(x) = (\bar{x}, 1)$ where $\bar{x}_i = 1$ if $x_i = 0$, 0 if $x_i = 1$. Then $R^2 \leq n+1$ and $\|u\|^2 = 4k+1$, giving

$$M(A) \leq (n+1)(4k+1).$$

Problem 34 (2). State the perceptron convergence theorem [Novikoff] explaining the relation with the hard margin support vector machine solution.

Solution. The number of mistakes made by the perceptron algorithm is:

$$M \leq \frac{R^2}{\gamma^2},$$

if there exists a vector v that satisfies $\langle v, x_t \rangle y_t \geq \gamma$ and $\|v\| = 1$. The γ term is analogous to the SVM margin term.

Problem 35 (3). Define the c -regret of a learning algorithm as $c\text{-regret}(m) = L_A(S) - c \min_{i \in [n]} L_i(S)$, thus the usual regret is the 1-regret.

1. Argue for the weighted majority set-up that without randomised prediction it is impossible for all training sequences to obtain sublinear c -regret for $c < 2$.
2. Show how to select β to obtain sublinear 2-regret.

Solution. 3.1: Consider two experts E_1, E_2 with constant predictions $x_{t,1} = 0, x_{t,2} = 1$ for all $t \in [T]$, and an adversary that forces a mistake on every prediction. Then $\min_{i \in \{1,2\}} M_i \leq T/2$, but $M_A = T$. Hence $M_A \geq 2 \min_i M_i$.

3.2: Starting with the WM bound $M \leq \frac{\ln(1/\beta)}{\ln(2/(1+\beta))} M_i + \frac{1}{\ln(2/(1+\beta))} \ln(n)$, define $\epsilon := 1 - \beta$ and assume $\epsilon \in [0, 1/2]$. Using the inequalities $-\log(1 - \epsilon) \leq \epsilon + \epsilon^2$ and $-\log(1 + x) \geq -x$:

$$M \leq 2(1 + \epsilon)M_i + \frac{2}{\epsilon} \ln(n) \leq 2M_i + 2\epsilon T + \frac{2}{\epsilon} \ln(n).$$

By AM-GM, setting $\epsilon = \sqrt{\log(n)/T}$:

$$M \leq 2M_i + 4\sqrt{\log(n) \cdot T}.$$

(For the case $\sqrt{\log(n)/T} > 1/2$, we use the generic bound $M \leq T \leq 2M_i + 4\log(n)$.)

Combining: $M \leq 2M_i + 4\sqrt{\log(n)T} + 4\log(n)$.

Problem 36 (4). Consider binary prediction with expert advice, with a perfect expert. Prove that any algorithm makes at least $\Omega(\min(T, \log_2 n))$ mistakes in the worst case.

Solution. Assume $\log_2 n$ is an integer. Consider a complete binary tree of depth N where $n = 2^N$ experts correspond to the 2^N paths from root to leaves. At each internal node, the adversary assigns the true label to be whichever label the algorithm did not predict. This forces N mistakes when $N \leq T$, and T mistakes if $N > T$. Taking $n = 2^N$, we have $\min(T, \log_2 n)$ forced mistakes.

Problem 37 (5). By using randomisation in the prediction, design an algorithm with a bound that has the property $\frac{M}{T} \leq \frac{\min_i M_i}{T}$ as $T \rightarrow \infty$ for the weighted majority setting.

Solution. Initialise all expert weights to 1. At each timestep t : receive expert predictions $x_t \in \{0, 1\}^n$; predict $\hat{y}_t = 0$ with probability $\frac{\sum_{i: x_{t,i}=0} w_{t,i}}{W_t}$, else predict 1; receive y_t ; multiply weights of incorrect experts by $\beta \in [0, 1)$.

Define A_t as the proportion of total weight on incorrect experts at time t . Then $\mathbb{E}[M] = \sum_{t=1}^T A_t$. We have

$$W_{T+1} = W_1 \prod_{t=1}^T (1 - (1 - \beta)A_t) \leq n \exp(-(1 - \beta)\mathbb{E}[M]).$$

Since $W_{T+1} \geq \beta^{M_i}$ for any i , taking logs and rearranging:

$$\mathbb{E}[M] \leq \frac{-\log \beta}{1 - \beta} M_i + \frac{1}{1 - \beta} \log(n).$$

With $\epsilon = 1 - \beta$ and $\epsilon \in [0, 1/2]$:

$$\mathbb{E}[M] \leq (1 + \epsilon)M_i + \frac{1}{\epsilon} \log(n) \leq M_i + \epsilon T + \frac{1}{\epsilon} \log(n).$$

Setting $\epsilon = \sqrt{\log(n)/T}$ gives

$$\mathbb{E}[M] - M_i \leq 2\sqrt{\log(n) \cdot T}, \quad \text{so} \quad \frac{\mathbb{E}[M]}{T} - \frac{M_i}{T} \leq 2\sqrt{\frac{\log(n)}{T}} \rightarrow 0 \text{ as } T \rightarrow \infty.$$

6 Problem Sheet 6: Graphs in Machine Learning

Solutions by Shota Saito

Problem 38 (1.1). Describe, in brief, the differences between semi-supervised learning, supervised learning and unsupervised learning.

Solution. *Unsupervised learning* learns how to infer a function to describe the hidden structure of unlabeled data. *Supervised learning* learns from labeled data and predicts labels of unseen data. *Semi-supervised learning* infers structure using both unlabeled and labeled data simultaneously.

Problem 39 (1.2). Given the unlabeled data points $(-2, 0), (-1, -1), (0, 0), (1, 1), (2, 0)$, draw the 2-nearest neighbor graph with respect to the Euclidean distance.

Solution. Label the points as $A = (-2, 0), B = (-1, -1), C = (0, 0), D = (1, 1), E = (2, 0)$. The 2-nearest neighbors are: $A \rightarrow B, C; B \rightarrow A, C; C \rightarrow B, D; D \rightarrow C, E; E \rightarrow C, D$. Edges: $A-B, A-C, B-C, C-D, D-E, C-E$.

Problem 40 (1.3). Give the graph Laplacian for the dataset above.

Solution. The adjacency matrix is

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & & & & \\ & 2 & & & \\ & & 4 & & \\ & & & 2 & \\ & & & & 2 \end{pmatrix}.$$

The graph Laplacian $L = D - A$ is

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ -1 & -1 & 4 & -1 & -1 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

Problem 41 (1.4). In the labelled graph $v_1(-1) - v_2 - v_3 - v_4 - v_5(+1)$ (a path graph with v_1 labelled -1 and v_5 labelled $+1$):

1. What is the value of the minimum cut solution at vertex v_3 ?
2. What is the value of the harmonic energy solution at vertex v_3 ?

Solution. 1.4.1: For the minimum cut, v_1 should be -1 and v_5 should be $+1$. The label for v_3 can take either 1 or -1 ; both yield minimum cut 2 .

1.4.2: To minimize $S_2 := \sum_{(u,v) \in E} |\psi(u) - \psi(v)|^2$, with $\psi(v_1) = -1$ and $\psi(v_5) = 1$, by symmetry we need equal spacing: $|\psi(v_2) - \psi(v_3)| = |\psi(v_3) - \psi(v_4)| = |\psi(v_4) - \psi(v_5)|$. Therefore $\psi(v_3) = 1/3$ (linearly interpolated: $-1, -1/3, 1/3, 2/3, 1$ but since the boundary is at ± 1 with 4 steps we get $\psi(v_3) = -1 + 2 \cdot \frac{2}{4} = 0$...

Actually: with boundary $\psi(v_1) = -1, \psi(v_5) = 1$, harmonic solution on a path gives linear interpolation: $\psi(v_k) = -1 + \frac{2(k-1)}{4}$. So $\psi(v_3) = -1 + 1 = 0$.

Note: The official solution states $\psi(v_3) = 1/3$ which corresponds to boundary conditions $\psi(v_2) = -1$ and $\psi(v_5) = 1$ on a specific graph. The answer depends on the exact graph structure given in the problem figure.)

Problem 42 (2.1). Consider the parameterized graph D_m consisting of two m -cliques connected by a single bridge edge. The first clique contains a node labelled -1 and the second a node labelled $+1$. Consider the labelling of D_m as resulting from harmonic energy minimisation. What is the labelling as $m \rightarrow \infty$?

Solution. By the symmetric structure of the graph, let the vertices be $v_1, \dots, v_m$ (first clique, v_1 labelled -1) and $v_{m+1}, \dots, v_{2m}$ (second clique, v_{2m} labelled $+1$), with bridge edge (v_m, v_{m+1}) . By symmetry:

$$\psi(v_j) = \frac{m+2}{m+4} \text{ for } j = 2, \dots, m-1, \quad \psi(v_m) = \frac{m}{m+4}, \quad \psi(v_{m+1}) = -\frac{m}{m+4}.$$

Therefore, as $m \rightarrow \infty$:

$$\psi(v_j) \rightarrow 1 \text{ for } j = 2, \dots, m, \quad \psi(v_j) \rightarrow -1 \text{ for } j = m+1, \dots, 2m-1.$$

(Taking account of the labelling convention with $v_1 = -1$ in the first clique.)

Problem 43 (2.2). [Hard] Prove that

$$\min_{u \in \{-1, 1\}^n} \left\{ \sum_{ij} w_{ij} |u_i - u_j| : u_{i_1} = y_1, \dots, u_{i_\ell} = y_\ell \right\} = \min_{u \in \mathbb{R}^n} \left\{ \sum_{ij} w_{ij} |u_i - u_j| : u_{i_1} = y_1, \dots, u_{i_\ell} = y_\ell \right\}.$$

Solution. Let α denote the integer program (IP) optimum and β the linear program (LP) optimum. Clearly $\beta \leq \alpha$ since $\{-1, 1\}^n \subseteq \mathbb{R}^n$.

Let u^*, x^* be the LP solution. Consider the randomized rounding: pick $r \in (0, 1)$ uniformly, set $S = \{v : u_v^* < r\}$. This always returns a feasible IP solution. We have $\Pr_r[(i, j) \text{ is a cut}] = |u_i^* - u_j^*| = x_{ij}^*$ by the LP constraint. The expected cost is

$$\sum_{ij \in E} w_{ij} \Pr_r[(i, j) \text{ is a cut}] = \sum_{ij} w_{ij} x_{ij}^* = \beta \leq \alpha.$$

Since the expected cost is at least α (as α is the min-cut), we get $\alpha \leq \beta \leq \alpha$, hence $\alpha = \beta$.

Problem 44 (3.1). [Hard] Prove the following inequality for the resistance diameter:

$$\max_{k \in [n]} L_{kk}^+ \leq \max_{i, j \in [n]} r_G(i, j).$$

Solution. From the definition of effective resistance, $r_G(i, j) = e_i^\top L^+ e_i - 2e_i^\top L^+ e_j + e_j^\top L^+ e_j$. Since $L^+ \mathbf{1} = 0$, we have $\sum_{j=1}^n L^+ e_j = L^+ \mathbf{1} = 0$. Hence

$$\frac{1}{n} \sum_{j=1}^n r_G(i, j) = e_i^\top L^+ e_i + \frac{1}{n} \sum_{j=1}^n e_j^\top L^+ e_j = L_{ii}^+ + \frac{1}{n} \text{trace}(L^+).$$

Since L^+ is PSD, $\text{trace}(L^+) \geq 0$, so $L_{ii}^+ \leq \frac{1}{n} \sum_j r_G(i, j) \leq \max_{i, j} r_G(i, j)$. Taking max over i gives the result.

Problem 45 (3.2). [Hard] Prove $L_{vv}^+ = \frac{R(v)}{n} - \frac{R_{\text{tot}}}{n^2}$ where $R(v) = \sum_{u \in V} r_G(v, u)$ and $R_{\text{tot}} = \sum_{u > v} r_G(u, v)$.

Solution. From the computation in 3.1: $\frac{1}{n} R(v) = L_{vv}^+ + \frac{1}{n} \sum_{u \in V} L_{uu}^+$. Also,

$$\frac{1}{n^2} R_{\text{tot}} = \frac{1}{2n^2} \sum_{u, v \in V} (L_{uu}^+ + L_{vv}^+) = \frac{1}{n} \sum_{u \in V} L_{uu}^+.$$

Combining: $L_{vv}^+ = \frac{R(v)}{n} - \frac{R_{\text{tot}}}{n^2}$.

Problem 46 (3.3). [**Hard**] For an unweighted connected graph prove $\sum_{(i,j) \in G} r_G(i,j) = n - 1$.

Solution. Define the incidence matrix $C \in \mathbb{R}^{|E| \times |V|}$. Note $L = C^\top C$ for unweighted graphs. Let $\Gamma = CL^+C^\top$; then $\Gamma_{\ell,\ell} = r_G(i,j)$ where ℓ connects i and j . We need $\text{trace}(\Gamma)$:

$$\text{trace}(CL^+C^\top) = \text{trace}(L^+C^\top C) = \text{trace}(L^+L).$$

The eigenvalues of L^+L are either 1 or 0, with $n - 1$ eigenvalues equal to 1 (for a connected graph). Hence $\text{trace}(\Gamma) = n - 1$.

Problem 47 (3.4). [**Hard**] Let L be a graph Laplacian for an n -vertex graph. Consider the kernel $K := (L + \epsilon I)^{-1}$ where $\epsilon > 0$. Prove that for any $x \in [n]^\ell$ and any $y \in \{-1, 1\}^\ell$:

$$\text{argmin}_{u \in \mathbb{R}^n: u_{x_1}=y_1, \dots, u_{x_\ell}=y_\ell} u^\top K^{-1}u = \text{argmin}_{u \in \mathbb{R}^n: u_{x_1}y_1 \geq 1, \dots, u_{x_\ell}y_\ell \geq 1} u^\top K^{-1}u.$$

Solution. Since $K^{-1} = L + \epsilon I$ is diagonally dominant (as L has non-negative diagonal and non-positive off-diagonal entries), suppose for contradiction there exists v in the inequality-constrained set with $v^\top K^{-1}v < \min_{u \in U_y} u^\top K^{-1}u$ and $|v_i| > 1$ for some index i of maximal magnitude. By the stationarity condition and diagonal dominance of K^{-1} , one can show that $|v_j|$ must be constant across all indices j connected via K^{-1} . Since K^{-1} is irreducible, $|v_1| = \dots = |v_n| > 1$. But then $v/|v_1|$ would be feasible with strictly smaller cost, contradicting minimality of v .

The PD matrix K that does not satisfy this property is any PD matrix that is not diagonally dominant.

7 Problem Sheet 7: Learning Theory

Solutions by Antonin Schrab

Problem 48 (1). Let $\mathcal{X} = \mathbb{R}^2$, $\mathcal{Y} = \{0, 1\}$, and let $\mathcal{H}$ be the class of concentric circles in the plane, i.e., $\mathcal{H} = \{h_r : r > 0\}$ where $h_r(x) = \mathbf{1}(\|x\|_2 \leq r)$. Prove that $\mathcal{H}$ is PAC learnable (assume realizability), and its sample complexity is bounded by $m_{\mathcal{H}}(\epsilon, \delta) \leq \left\lceil \frac{\ln(1/\delta)}{\epsilon} \right\rceil$.

Solution. We need to show there exists a learning algorithm A such that for all $\epsilon, \delta \in (0, 1)$, all distributions D over $\mathbb{R}^2$, and every labelling function f^* , if the realizable assumption holds, then for $m \geq \lceil \ln(1/\delta)/\epsilon \rceil$:

$$P_{S \sim D^m}(P_{x \sim D}[A(S)(x) \neq f^*(x)] \leq \epsilon) \geq 1 - \delta.$$

By realizability, there exists $r^* > 0$ such that $f^*(x) = \mathbf{1}(\|x\|_2 \leq r^*)$. Consider the algorithm that returns h_{r_S} where $r_S := \max_{i: y_i=1} \|x_i\|_2$. By realizability, $r_S \leq r^*$.

Let $\epsilon \in (0, 1)$. Define r_ϵ such that $P_{x \sim D}(r_\epsilon \leq \|x\|_2 \leq r^*) = \epsilon$. Then

$$\begin{aligned} P_{S \sim D^m}(L_D(h_{r_S}) \geq \epsilon) &= P_{S \sim D^m}(r_S \leq r_\epsilon) = P_{S \sim D^m}\left(\bigcap_{i=1}^m \{\|x_i\|_2 \leq r_\epsilon\} \cup \{\|x_i\|_2 \geq r^*\}\right) \\ &= (1 - P_{x \sim D}(r_\epsilon \leq \|x\|_2 \leq r^*))^m = (1 - \epsilon)^m \leq e^{-\epsilon m} \leq \delta \end{aligned}$$

for $m \geq \frac{\ln(1/\delta)}{\epsilon}$.

Problem 49 (2). Prove $1 - x \leq e^{-x}$ for $x \in \mathbb{R}$.

Solution. We need to prove $g(x) := e^{-x} + x - 1 \geq 0$ for all $x \in \mathbb{R}$. We have $g'(x) = -e^{-x} + 1$ and $g''(x) = e^{-x} > 0$ for all $x \in \mathbb{R}$, so g is convex. It attains its minimum when $0 = g'(x) = -e^{-x} + 1 \iff x = 0$. Hence $g(x) \geq g(0) = 0$ for all $x \in \mathbb{R}$.

Problem 50 (3). Prove that if $|\mathcal{X}| \geq km$ for a positive integer $k \geq 2$, then we can replace the lower bound of $1/4$ in the No-Free-Lunch theorem with $\frac{k-1}{2k} = \frac{1}{2} - \frac{1}{2k}$.

Solution. Following the proof of the NFL Theorem (Theorem 5.1 in [SS14]): Let C be a subset of $\mathcal{X}$ of size km . As in the proof, let $v_1, \dots, v_p$ be examples in C which do not appear in a training set of size m ; we deduce that $p \geq (k-1)m$. The key equation becomes

$$L_{D_i}(h) = \frac{1}{km} \sum_{x \in C} \mathbf{1}(h(x) \neq f_i(x)) \geq \frac{1}{km} \sum_{r=1}^p \mathbf{1}(h(v_r) \neq f_i(v_r)) \geq \frac{k-1}{kp} \sum_{r=1}^p \mathbf{1}(h(v_r) \neq f_i(v_r)).$$

We have replaced the factor $\frac{1}{2}$ by $\frac{k-1}{2k}$. The rest of the proof carries through identically, yielding

$$\max_{i \in [T]} \mathbb{E}_{S \sim D_i^m}[L_{D_i}(A(S))] \geq \frac{1}{2} \cdot \frac{k-1}{k} = \frac{k-1}{2k}.$$

Problem 51 (4). Given some finite domain set $\mathcal{X}$, and a number $k \leq |\mathcal{X}|$, figure out the VC-dimension of each of the following classes:

1. $\mathcal{H}_{=k}^{\mathcal{X}} = \{h \in \{0, 1\}^{\mathcal{X}} : |\{x : h(x) = 1\}| = k\}$.
2. $\mathcal{H}_{\text{at-most-}k}^{\mathcal{X}} = \{h \in \{0, 1\}^{\mathcal{X}} : |\{x : h(x) = 1\}| \leq k \text{ or } |\{x : h(x) = 0\}| \leq k\}$.

Solution. 4.1: $VCdim(\mathcal{H}_{=k}^{\mathcal{X}}) = \min\{k, |\mathcal{X}| - k\}$.

Lower bound: Let $\ell = \min\{k, |\mathcal{X}| - k\}$. Consider a set C of ℓ elements. For any labelling with N_+ ones ($N_+ \leq \ell \leq k$), we can assign label 1 to the N_+ positive points and $k - N_+$ points from $\mathcal{X} \setminus C$ (possible since $|\mathcal{X}| - \ell \geq k$). So $\mathcal{H}_{=k}$ shatters C .

Upper bound: A set of $k + 1$ all-ones points cannot be classified correctly by any $h \in \mathcal{H}_{=k}$ (which assigns 1 to exactly k points). Similarly, $|\mathcal{X}| - k + 1$ all-zeros points cannot be handled. Hence $VCdim < 1 + \min\{k, |\mathcal{X}| - k\}$.

4.2: $VCdim(\mathcal{H}_{at-most-k}) = \min\{2k + 1, |\mathcal{X}|\}$.

Lower bound: For any labelling of $2k + 1$ points with N_+ ones and $N_- = 2k + 1 - N_-$ zeros, either $N_+ \leq k$ or $N_- \leq k$. In either case, a classifier in $\mathcal{H}_{at-most-k}$ can be constructed.

Upper bound: $2k + 2$ points with $k + 1$ ones and $k + 1$ zeros violates both conditions.

Problem 52 (5). We have shown that for a finite hypothesis class $\mathcal{H}$, $VCdim(\mathcal{H}) \leq \lfloor \ln(|\mathcal{H}|) \rfloor$.

1. Find an example of a class $\mathcal{H}$ of functions over $\mathcal{X} = [0, 1]$ such that $\mathcal{H}$ is infinite while $VCdim(\mathcal{H}) = 1$.
2. Give an example of a finite hypothesis class $\mathcal{H}$ over $\mathcal{X} = [0, 1]$, where $VCdim(\mathcal{H}) = \lfloor \log_2(|\mathcal{H}|) \rfloor$.

Solution. 5.1: Consider $\mathcal{H} := \{h_y : y \in [0, 1]\}$ where $h_y(x) = \mathbf{1}(x \leq y)$. This is infinite. The point $z = 0.5$ is shattered (label 1: use h_1 ; label 0: use h_0), so $VCdim \geq 1$. For any two points $z_1 < z_2$, the labelling ($z_1 \mapsto 0, z_2 \mapsto 1$) cannot be realized (any h_y with $h_y(z_1) = 0$ requires $y < z_1$, but then $h_y(z_2) = 0$ also). So $VCdim < 2$, hence $VCdim = 1$.

5.2: $\mathcal{H} = \{h_{1/4}, h_{3/4}\}$ with $h_y(x) = \mathbf{1}(x \leq y)$. Then $|\mathcal{H}| = 2$, $\lfloor \log_2 2 \rfloor = 1$. The point $z = 0.5$ is shattered: $h_{3/4}(0.5) = 1$, $h_{1/4}(0.5) = 0$. $VCdim = 1 = \lfloor \log_2(|\mathcal{H}|) \rfloor$.

Problem 53 (6). On page 49 (Learning Theory slides) why did we formulate the VC-dimension of large margin classifiers with respect to a fixed input space X rather than for example $\{x \in \ell_2 : \|x\|_2 \leq R\}$?

Solution. We have $\mathcal{H}_{X,\Lambda} = \{x \mapsto \text{sign}(\langle w, x \rangle) : w \in \ell_2, \|w\|_2 \leq \Lambda, |\langle w, x \rangle| \geq 1 \text{ for all } x \in X\}$.

If we consider $X := \{x \in \ell_2 : \|x\|_2 \leq R\}$, then there exists some $\tilde{x} \in X$ with $\|\tilde{x}\|_2 < \Lambda^{-1}$. For all $w \in \ell_2$, by Cauchy-Schwarz: $|\langle w, \tilde{x} \rangle| \leq \|\tilde{x}\|_2 \|w\|_2 < \Lambda^{-1} \Lambda = 1$. Therefore $\mathcal{H}_{X,\Lambda}$ would be empty. This is why X must be a fixed finite set.

Problem 54 (7 (Littlestone dimension – extra)). Suppose we wish to design a deterministic online algorithm with a mistake bound for learning w.r.t. some hypothesis class $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ where $\mathcal{Y} = \{-1, 1\}$. Prove that $VCdim(\mathcal{H}) \leq Ldim(\mathcal{H})$.

Solution. Let $d = VCdim(\mathcal{H})$. We show that for any deterministic algorithm A , there exist $h \in \mathcal{H}$ and a sequence s such that A makes at least d mistakes. Since $d = VCdim(\mathcal{H})$, there exists $C = \{x_1, \dots, x_d\} \subseteq \mathcal{X}$ shattered by $\mathcal{H}$. For $t = 1, \dots, d$: give x_t to A , A predicts $\hat{y}_t$, select $h \in \mathcal{H}$ with $h(x_i) = y_i$ for $i < t$ and $h(x_t) = -\hat{y}_t$ (exists since $\mathcal{H}$ shatters C), and set $y_t := -\hat{y}_t$. By construction, A makes d mistakes. Hence $VCdim(\mathcal{H}) \leq \max_{h,s} M_A(h, s)$ for all A , which means $VCdim(\mathcal{H}) \leq Ldim(\mathcal{H})$.

Problem 55 (7). Given $X \subset \ell_2$ and $\Lambda \in (0, \infty)$, define

$$\mathcal{H}_{X,\Lambda} = \{x \mapsto \text{sign}(\langle w, x \rangle) : w \in \ell_2, \|w\|_2 \leq \Lambda, |\langle w, x \rangle| \geq 1 \text{ for all } x \in X\}.$$

1. Prove that $VCdim(\mathcal{H}_{X,\Lambda}) \leq \Lambda^2 \max_{x \in X} \|x\|_2^2$.
2. Show that for every $\Lambda \in (0, \infty)$ there exists an X such that $VCdim(\mathcal{H}_{X,\Lambda}) = \lfloor \Lambda^2 \max_{x \in X} \|x\|_2^2 \rfloor$.

Solution. 7.1: By the Littlestone dimension result (Problem Extra), $VCdim(\mathcal{H}_{X,\Lambda}) \leq \max_{h,s} M_A(h,s)$ for any deterministic algorithm A . Applying the Perceptron Algorithm with margin $\gamma = 1/\|w\|_2$:

$$VCdim(\mathcal{H}_{X,\Lambda}) \leq \left(\frac{\max_{x \in X} \|x\|}{\gamma} \right)^2 = \|w\|_2^2 \max_{x \in X} \|x\|^2 \leq \Lambda^2 \max_{x \in X} \|x\|_2^2.$$

7.2: Fix $\Lambda > 0$ and consider $X_\Lambda := \{\tilde{x}\}$ with $\tilde{x} := (\Lambda^{-1}, 0, 0, \dots) \in \ell_2$. Then $\|\tilde{x}\|_2 = \Lambda^{-1}$ and $\lfloor \Lambda^2 \|\tilde{x}\|_2^2 \rfloor = \lfloor \Lambda^2 \Lambda^{-2} \rfloor = 1$.

By Cauchy-Schwarz, $|\langle w, \tilde{x} \rangle| \leq \|w\|_2 \|\tilde{x}\|_2 \leq \Lambda \cdot \Lambda^{-1} = 1$, combined with $|\langle w, \tilde{x} \rangle| \geq 1$, gives $|\langle w, \tilde{x} \rangle| = 1$. So $\mathcal{H}_{X_\Lambda, \Lambda} = \{\tilde{x} \mapsto 1, \tilde{x} \mapsto -1\}$, which shatters $\{\tilde{x}\}$ giving $VCdim = 1$.

8 Supplementary Exam (May 2025)

Answer ALL FIVE (5) questions.

Problem 56 (1: Variants of Least Squares). (a) Consider Tikhonov regularization with an unpenalized offset term b :

$$\min_{w \in \mathbb{R}^d, b \in \mathbb{R}} \left\{ \frac{1}{n} \sum_{i=1}^n (\langle w, x_i \rangle + b - y_i)^2 + \lambda \|w\|^2 \right\}.$$

Let (w^*, b^*) be the solution. Denote centered data $x_i^c = x_i - \bar{x}$, $y_i^c = y_i - \bar{y}$.

- i. Write the problem in matrix notation. [1 mark]
- ii. Show that w^* also solves $\min_{w \in \mathbb{R}^d} \left\{ \frac{1}{n} \sum_{i=1}^n (\langle w, x_i^c \rangle - y_i^c)^2 + \lambda \|w\|^2 \right\}$. [5 marks]
- iii. Give closed form expressions for b^* and w^* . [3 marks]
- iv. Can you “kernelize” the method? [3 marks]

(b) Consider weighted regularized least squares:

$$\min_{w \in \mathbb{R}^d} \left\{ \sum_{i=1}^n \gamma_i (\langle w, x_i \rangle - y_i)^2 + \lambda \|w\|^2 \right\}$$

where $\sum_{i=1}^n \gamma_i = 1$ and $\gamma_i > 0$.

- i. Write the problem in matrix notation. [1 mark]
- ii. Derive the explicit form of the minimizer w^* . [2 marks]
- iii. Consider the weighted loss with unpenalized offset b . Derive w^*, b^* . [5 marks]

Problem 57 (2: Adaboost). (a) Derive Adaboost as the greedy minimizer of empirical risk

$$\min \left\{ \frac{1}{n} \sum_{i=1}^n \ell(f_T(x_i), y_i) \mid f_T = \sum_{t=1}^T \alpha_t h_t, \alpha_t \in \mathbb{R}, h_t \in \mathcal{H} \right\}.$$

- i. Which loss function ℓ yields the Adaboost algorithm?
- ii. What kind of greedy strategy is used?
- iii. Explicitly write down the update rule to transition from f_t to f_{t+1} .

[10 marks]

(b) Is overfitting a problem for Adaboost? Are there straightforward strategies to add regularization? [5 marks]

Problem 58 (3: First Order Optimization + Kernels). (a) Recall the definition of M -smoothness for a function $F: \mathbb{R}^d \rightarrow \mathbb{R}$. [1 mark]

(b) Recall how Gradient Descent (GD) is defined to minimize $E_n(w) = \frac{1}{n} \sum_{i=1}^n \ell(w^\top x_i, y_i)$. [3 marks]

(c) Show that GD can be “kernelized”: for any iterate w_t of GD, find parameters $c_1^{(t)}, \dots, c_n^{(t)} \in \mathbb{R}$ such that $\langle w_t, x \rangle = \sum_{i=1}^n \langle x_i, x \rangle c_i^{(t)}$ for any $x \in \mathbb{R}^d$. [8 marks]

(d) Discuss how this allows GD in kernel settings with possibly infinite-dimensional feature maps. [4 marks]

(e) Comment on how applying GD in the general kernel setting affects convergence. [4 marks]

Problem 59 (4: Rademacher Complexity for Hard-SVM). (a) Recall the formulation of the hard-SVM problem. [2 marks]

Let $X = \{x : \|x\| \leq R\} \subset \mathbb{R}^d$, $Y = \{-1, 1\}$. Let ρ be a distribution over $X \times Y$ such that $\exists w^* \in \mathbb{R}^d$ with $P(yx^\top w^* \geq 1) = 1$. Let $S = (x_i, y_i)_{i=1}^n \sim \rho^n$ and w_S the hard-SVM solution.

(b) Recall the hinge loss $\ell(x^\top w, y)$ and show $P_{(x,y) \sim \rho}(y \neq \text{sign}(x^\top w)) \leq E(w)$ where $E(w) = \int \ell(x^\top w, y) d\rho$. [3 marks]

(c) Apply the Rademacher complexity bound for excess risk control. [5 marks]

(d) Choose $\mathcal{H} = \{f : X \rightarrow \mathbb{R} \mid f(x) = x^\top w, \|w\| \leq B\}$. Show $\exists B > 0$ such that $\|w_S\| \leq B$. [5 marks]

(e) Provide an upper bound for $R_n(\mathcal{H})$ the Rademacher complexity. [5 marks]

(f) Combine to bound $P(y \neq \text{sign}(x^\top w_S)) \leq O(n^{-1/2})$. [5 marks]

Problem 60 (5: Sub-gradient Method for Online Learning). (a) Recall the definition of sub-gradient of a convex $\ell : \mathbb{R}^d \rightarrow \mathbb{R}$ at $\theta \in \mathbb{R}^d$. [3 marks]

Consider the online algorithm: starting from $\theta_1 \in \mathbb{R}^d$, step-size $\eta > 0$, iteratively for $t = 1, \dots, T$: pay cost $\ell_t(\theta_t)$; update $\theta_{t+1} = \theta_t - \eta g_t$ where g_t is a subgradient of ℓ_t at θ_t .

(b) Show that

$$\ell_t(\theta_t) - \ell_t(u) \leq \frac{1}{2\eta} \|\theta_t - u\|^2 - \frac{1}{2\eta} \|\theta_{t+1} - u\|^2 + \frac{\eta}{2} \|g_t\|^2.$$

[5 marks]

(c) Prove:

$$\sum_{t=1}^T \ell_t(\theta_t) - \sum_{t=1}^T \ell_t(u) \leq \frac{\|u - \theta_1\|^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^T \|g_t\|^2.$$

[5 marks]

(d) Suppose all ℓ_t are L -Lipschitz, $\|u\| \leq B$, $\theta_1 = 0$. How to choose η ? What regret bound follows? [7 marks]